Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 29)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

0 1 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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0 1 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

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0 1 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

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0 1 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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0 2 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 2 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 3 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 3 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 3 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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0 4 . 1 Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 4 . 2 Describe **five** features of a suitable release site.

[5 marks]

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0 4 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 5 . 1 Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

0 5 . 2 Explain **four** reasons why mangroves are important.

[4 marks]

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0 5 . 3 Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

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0 6 Table 1 lists some techniques used in ecological monitoring.

0 6 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

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0 6 . 2 Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

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0 6 . 3 Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

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0 7 . 1 Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

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0 7 . 2 Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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0 7 . 3 Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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0 8 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 8 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 8 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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0 9 . 1 Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

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0 9 . 2 Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

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0 9 . 3 Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

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0 9 . 4 Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

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1 0 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

1 0 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

1 0 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

1 0 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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Write an essay on one of the following topics.

11 . 1 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

OR

11 . 2 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS